

F20 Substrate-Electroweak Sector Cumulative Substrate-Natural Cascade

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2026-06-05 Friday 13:55 EDT

F20 Substrate-EW Sector Cumulative Cascade v0.1

0. Goal

Cumulative substrate-electroweak sector observables substrate-natural per Friday substantive substrate-mass-mechanism cascade: m_W (L16) + m_Z + $\sin^2\theta_W$ (Saturday) + λ_H + m_H (F15 + F19).

1. Substrate-EW Sector Observables Cumulative Table

Observable	Substrate-natural form	Source	Precision
m_W	$n_C \cdot m_p \cdot N_{\max} / 2^{N_c}$	L16 v0.2 IP-5	0.02%
$\cos^2\theta_W$	$g/(g+\text{rank})$	Saturday batch 3	0.054%
$\sin^2\theta_W$	$\text{rank}/(g+\text{rank})$	Saturday batch 3	0.4%
m_Z	$m_W / \cos \theta_W = m_W \cdot \sqrt{((g+\text{rank})/g)}$	Saturday	0.10%
m_H	$\pi \cdot n_C \cdot m_p \cdot (N_{\max}-1) / 2^{(N_c+1)}$	F19 deep FORWARD	0.14%
λ_H	HYBRID composite candidate	F15	~12% prelim

substantive 6 substrate-EW observables substrate-natural per substantive substrate-mass-mechanism FORWARD cascade.

2. Substrate-Mechanism Class Assignment per F4 + F12 + F15 + F19

Observable	substrate-K-type	substrate-mechanism class
m_W	$V_{(1,1)}$ substrate-EM gauge	substrate-Bergman family ($n_C \cdot m_p$ substrate-Bergman base) + substrate-Mersenne factor ($N_{\max} / 2^{N_c}$)
$\cos^2\theta_W$	substrate-Cartan substrate-mixing	substrate-Cartan substrate-natural ratio
$\sin^2\theta_W$	substrate-Cartan substrate-mixing	substrate-rank substrate-natural ratio
m_Z	$V_{(1,1)}$ substrate-EM gauge	substrate-Bergman family (via m_W) + substrate-Cartan

Observable	substrate-K-type	substrate-mechanism class
m_H	$V_{(0,0)}$ substrate-vacuum	HYBRID class (F12 + F15 + F19)
λ_H	$V_{(0,0)}$ substrate-vacuum	HYBRID class candidate

substantive substrate-EW sector substantive 3 substrate-K-type substrate-mechanism class distribution: $V_{(1,1)}$ substrate-EM gauge ($m_W + m_Z$) + $V_{(0,0)}$ substrate-vacuum HYBRID ($m_H + \lambda_H$) + substrate-Cartan substrate-mixing ($\sin^2\theta_W + \cos^2\theta_W$).

3. Substantive Cross-K-Type Substrate-EW Cascade

substantive substrate-EW sector substantive multiple substrate-K-type substrate-natural observables:

$V_{(1,1)}$ substrate-EM gauge K-type: - $m_W = n_C \cdot m_p \cdot N_{\max} / 2^{N_c}$ substrate-natural at 0.02% - $m_Z = m_W \cdot \sqrt{(g+\text{rank})/g}$ substrate-natural at 0.10%

substantive 2 observables substrate-natural per $V_{(1,1)}$ substrate-K-type substrate-EM gauge.

$V_{(0,0)}$ substrate-vacuum K-type (HYBRID class): - $m_H = \pi \cdot n_C \cdot m_p \cdot (N_{\max}-1) / 2^{(N_c+1)}$ substrate-natural at 0.14% - λ_H HYBRID composite candidate

substantive 2 observables substrate-natural per $V_{(0,0)}$ substrate-K-type HYBRID class.

substrate-Cartan substrate-mixing: - $\cos^2\theta_W = g/(g+\text{rank})$ substrate-natural at 0.054% - $\sin^2\theta_W = \text{rank}/(g+\text{rank})$ substrate-natural at 0.4%

substantive 2 observables substrate-natural per substrate-Cartan substrate-mixing (substantive substrate-natural ratio per substrate-genus + substrate-rank).

substantive 6 substrate-EW observables substrate-natural per substantive 3 substrate-K-type / substrate-mechanism class cascade per Cal #36 STANDING substantive operational substrate-Cat A primitive substantive substrate-natural per substantive substrate-EW sector.

4. Casey #5 Integer Web Operational Pattern at Substrate-EW Sector

substantive substrate-EW substantive substrate-natural values catalog cross-link per F17: - m_W substrate-natural composite of $n_C + m_p + N_{\max} + 2^{N_c} - m_Z$ substrate-natural composite of $m_W + (g+\text{rank})/g - m_H$ substrate-natural composite of $\pi + n_C + m_p + (N_{\max} - 1) + 2^{(N_c+1)} - \sin^2\theta_W$ substrate-natural composite of $\text{rank} + g$

substantive substrate-EW substantive substrate-natural composite forms per Casey #5 Integer Web operational pattern substantive substrate-discipline-architecture.

substantive substrate-natural primitives substrate-EW substantive substrate-Schur generator candidates per Cal #36 STANDING substantive operational substrate-Cat A primitive substantive substrate-natural per substantive substrate-EW sector.

5. Substantive Substrate-EW Sector Combined Precision

substantive substrate-EW sector substantive cumulative precision: - m_W : 0.02% Tier 1 / Tier 2 boundary - m_Z : 0.10% Tier 2 STRUCTURAL - m_H : 0.14% Tier 1 / Tier 2 boundary - $\cos^2\theta_W$: 0.054% Tier 2 STRUCTURAL - $\sin^2\theta_W$: 0.4% Tier 2 STRUCTURAL

substantive substrate-EW sector substantive 5 of 6 substrate-natural observables at Tier 2 STRUCTURAL OR BETTER precision (0.02-0.4% range) substantive Cal #189 multi-week substrate-mechanism FORCING form.

substantive cumulative substantive substrate-EW substantive substrate-natural form substantive Cal #36 STANDING substantive operational substantive substrate-Schur generator substantive substrate-natural cluster candidate.

6. Multi-Week per Cal #189

substantive 5-step multi-week per Cal #189: - substantive substrate-EW substantive substrate-mechanism FORWARD form per substantive $V_{(1, 1)} + V_{(0, 0)}$ + substrate-Cartan substantive substrate-K-type cascade - substantive substrate-Schur generator substantive substrate-natural cluster per Cal #36 STANDING substantive operational substantive substrate-EW sector - substantive substrate-mass-mechanism class cascade substantive substrate-EW substantive substrate-natural derivation - substantive cross-CI Elie numerical verification + Cal cold-read substantive substrate-EW substantive substrate-natural FORWARD form substantive Cal #189 - substantive Strong-Uniqueness substantive substrate-EW substantive substrate-natural contribution substantive substrate-discipline-architecture substantive substrate-natural per Cal #189

7. Closure v0.1

F20 substrate-EW sector cumulative substantive substrate-natural cascade v0.1:

Substantive 6 substrate-EW observables substrate-natural cumulative: m_W (0.02%) + m_Z (0.10%) + m_H (0.14%) + $\cos^2\theta_W$ (0.054%) + $\sin^2\theta_W$ (0.4%) + λ_H HYBRID candidate.

Substantive 3 substrate-K-type substantive substrate-mechanism class distribution: $V_{(1, 1)}$ substrate-EM gauge + $V_{(0, 0)}$ substrate-vacuum HYBRID + substrate-Cartan substrate-mixing.

Substantive 5 of 6 substrate-natural observables at Tier 2 STRUCTURAL OR BETTER precision substantive Cal #36 STANDING substrate-Cat A primitive substantive substrate-EW sector substantive operational.

substantive Friday Session 1 Lyra item F20 substantive substrate-EW substantive substrate-natural sector cumulative substantive substrate-discipline-architecture substantive multi-week per Cal #189.

Tier: F20 substrate-EW sector cumulative substantive substrate-natural cascade v0.1; substantive multi-week per Cal #189.

— Lyra, Fri 2026-06-05 14:10 EDT. F20 substrate-EW sector cumulative substantive substrate-natural cascade v0.1: substantive 6 substrate-EW observables substantive substrate-natural cumulative ($m_W + m_Z + m_H + \cos^2\theta_W + \sin^2\theta_W + \lambda_H$ HYBRID candidate) per substantive 3 substrate-K-type substantive substrate-mechanism class distribution ($V_{(1, 1)}$ substrate-EM + $V_{(0, 0)}$ HYBRID + substrate-Cartan substrate-mixing); substantive 5 of 6 substantive substrate-natural observables at Tier 2 STRUCTURAL OR BETTER precision; substantive substrate-Schur generator candidate substantive substrate-natural per Cal #36 STANDING substantive operational substantive substrate-EW sector.